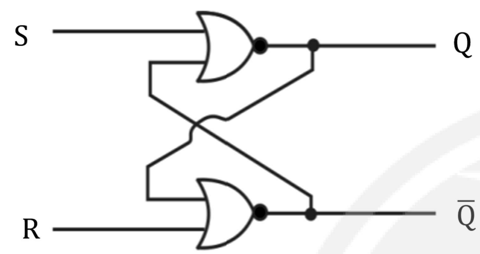
DPP 01

CS & IT

Digital Logic

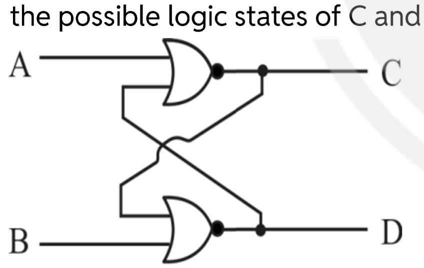
Miscellaneous Topics

Q1 In an SR latch made by cross-coupling two NAND gates, if both S and R inputs are set to 0, then it will result in



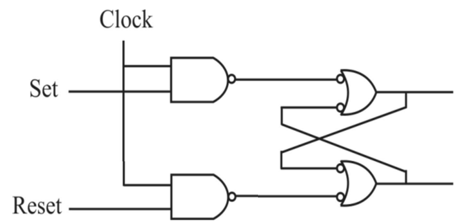
(A) Q = 0, Q' = 1
(B) Q = 1, Q' = 0
(C) Q = 1, Q' = 1
(D) Indeterminate states

Q2 In the circuit shown below, when inputs A = B = 0, the possible logic states of C and D are



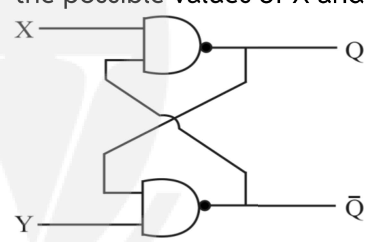
(A) C = 0, D = 1 or C = 1, D = 0
(B) C = 1, D = 0
(C) C = 1, D = 1 or C = 0, D = 0
(D) C = 0, D = 1

Q3 The two NAND gates before the latch circuit shown below, are used to



(A) act as buffers
(B) operate the latch faster
(C) avoid racing
(D) invert the latching action

Q4 In the circuit shown below, outputs $Q\overline{Q} = 01$, the possible values of X and Y are



(A) X = 1, Y = 0
(B) X = 1, Y = 1
(C) X = 0, Y = 1
(D) X = 0, Y = 0

Q5 Latch is a device with
(A) One stable state
(B) Two stable state
(C) Three stable state
(D) Infinite stable state

Q6 The two addition operations 24 + 14 = 41 and 23 + 12 = 101 are performed on number bases b_1 and

b_2 respectively. The values of b_1 and b_2 are respectively

(A) 7 and 4 (B) 4 and 7

(C) 8 and 4 (D) 4 and 8

Q7 If x and y are successive numbers in a number system of base b such that $(xy)_b = (25)_{10}$ and $(yx)_b = (31)_{10}$, then

(A) x = 4, y = 5 and b = 7

(B) x = 3, y = 4 and b = 6

(C) x = 4, y = 5 and b = 6

(D) x = 3, y = 4 and b = 7

Q8 If a = (4.4)5 and b = (3.3)5, then a + b = (x)5. The subscript 5 denotes the base on which the corresponding number is expressed. The value of x is

(A) 31.2 (B) 7.2

(C) 8.7 (D) 13.2

Q9 If $(X1CY)_{16} = (120702)_8$, then X and Y are

(A) A and 2 (B) B and 1

(C) 1 and B (D) 2 and A

Q10 Given $(135)_b + (144)_b = (323)_b$, where subscript b denotes the base on which numbers are expressed. What is value of b?

(A) 4 (B) 5

(C) 6 (D) 7

Answer Key

Q1	(C)	Q6	(A)
Q2	(A)	Q7	(D)
Q3	(D)	Q8	(D)
Q4	(A, B)	Q9	(A)
Q5	(B)	Q10	(C)

Hints & Solutions

Q1 Text Solution:

Q = 1, Q' = 1

Q2 Text Solution:

C = 0, D = 1 or C = 1, D = 0

Q3 Text Solution:

invert the latching action

Q4 Text Solution:

X = 1, Y = 0

X = 1, Y = 1

Q5 Text Solution:

Two stable state

Q6 Text Solution:

7 and 4

Q7 Text Solution:

x = 3, y = 4 and b = 7

Q8 Text Solution:

13.2

Q9 Text Solution:

A and 2

Q10 Text Solution:

6